

Lecture 12 (February 28)

The problem set is due today, but since a lot of people have asked for extensions, there will be a penalty-free grace period until Sunday. If you do submit on time, though, you might get some extra credit if you're e.g. on the cusp of A and A+.

In total, there will be "a few" problem sets in the semester. In past years, there were 3-4, but they were longer. This year we'll see but we'll likely have a problem set every couple of weeks.

Next week, Prof. Postnikov will be away at a conference, so Ilani and Dora (grad students) will be giving lectures.

Introducing rook placements

Today, we will talk about non-attacking rook placements, i.e. arrangements of rooks on a chessboard such that there is at most one rook in each row/column.

If you're placing n rooks on a $n \times n$ board, it's essentially the same as finding permutations, so there are $n!$ of them.

A generalization:

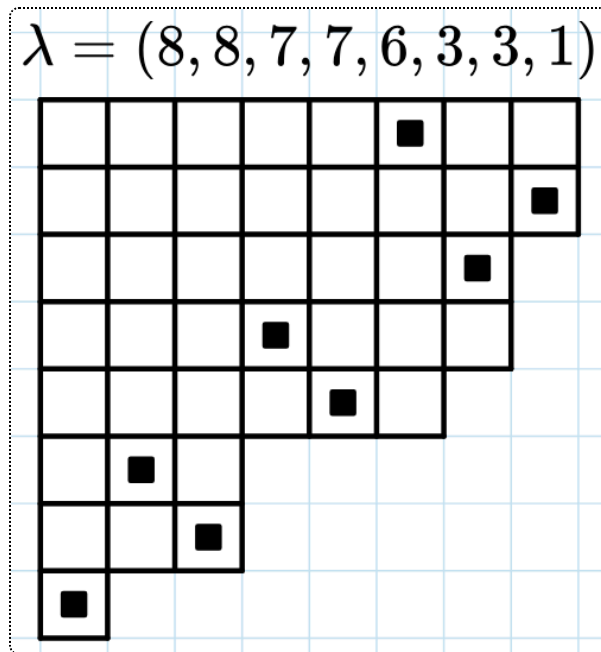
Theorem 1 (Rook placements on grids).

The number of ways to place k rooks on a $m \times n$ chessboard is

$$\binom{m}{k} \binom{n}{k} k!.$$

Indeed, you have to pick the k rows and the k columns on which the rooks will be placed, and then there are $k!$ ways to match the rows to the columns.

What if we want to place rooks on an arbitrary Young diagram, like this?



Theorem 2 (Rook placements on Young diagrams).

The number of ways to place n rooks on $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n)$ is equal to

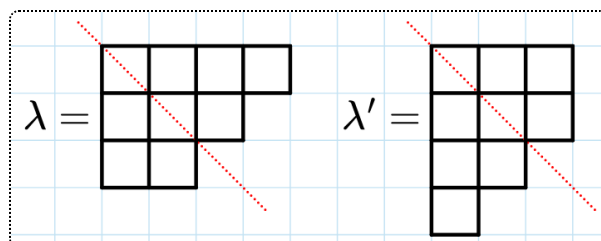
$$\prod_{i=1}^n (\lambda_i - (n - i)) = \lambda_n(\lambda_{n-1} - 1)(\lambda_{n-2} - 2) \cdots (\lambda_1 - n + 1)$$

Proof. There must be a rook on every row. We start from the bottom; there are λ_n choices. Then, for the second-to-last row, exactly one column is eliminated, so there are $\lambda_{n-1} - 1$ options. We continue in this way.

For our λ above, this is $1 \cdot 2 \cdot 1 \cdot 3 \cdot 3 \cdot 2 \cdot 2 \cdot 1 = 72$.

Assume $\lambda \subset n \times n$, i.e. it has at most n rows and n columns.

Young diagrams/partitions have an operation of "conjugation", essentially reflection over the main diagonal. For example:



We denote the conjugate $\lambda' = (\lambda'_1, \dots, \lambda'_n)$. As a corollary of Theorem 2, we have that

$$\begin{aligned} & \lambda_n(\lambda_{n-1} - 1)(\lambda_{n-2} - 2) \cdots (\lambda_1 - n + 1) \\ &= \lambda'_n(\lambda'_{n-1} - 1)(\lambda'_{n-2} - 2) \cdots (\lambda'_1 - n + 1) \end{aligned}$$

In our example, $\lambda' = (8, 7, 7, 5, 5, 5, 4, 2)$ and the product is $2 \cdot 3 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 1$.

In fact, we have a stronger claim:

Claim (Stronger claim about conjugates).

The n terms in the above products are exactly the same, just rearranged. In other words, the terms in the product with λ'_n correspond exactly to those in the other product.

Take some time to think about this claim (it may or may not appear on the next problem set.)

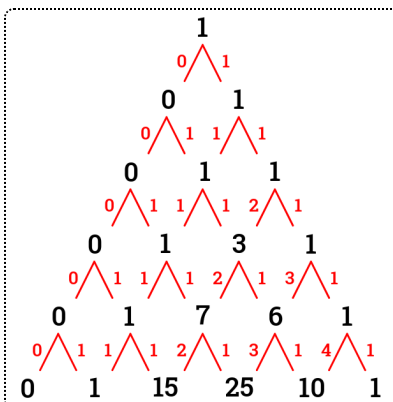
Connection to Stirling numbers

Recall that Stirling numbers of the second kind are

$$S(n, k) = \# \text{ set partitions of } [n] \text{ with } k \text{ blocks.}$$

There are also the associated **Bell numbers** $B(n)$, which are the total number of set-partitions of $[n]$. Essentially, $B(n) = \sum_k S(n, k)$.

Using the Stirling triangle of the second kind:

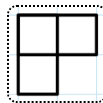


we can compute the first few Bell numbers.

n	$B(n)$
0	1
1	1
2	2
3	5
4	15
5	52

Let $\lambda = (n - 1, n - 2, n - 3, \dots, 1)$ be the "triangular" or "staircase" shape. How many ways can we place different number of rooks in this shape?

For $n = 3$, seen below, there is one way to place zero rooks, there are three ways to place two rooks, and there is one way to place two rooks.



For $n = 4$, after investigation, we see:

# rooks	# placements
0	1
1	6
2	$2 \cdot 2 + 1 \cdot 2 + 1 = 7$
3	1

These sure look like the Stirling numbers!

Theorem 3 (Placements in triangular shapes).

The number of ways to place any number of rooks (including 0) on the staircase shape $\lambda = (n - 1, n - 2, \dots, 1)$ equals B_n .

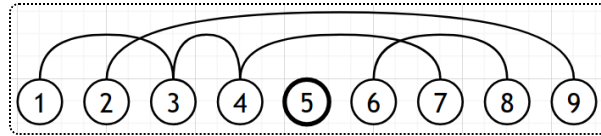
Moreover, if the number of rooks is fixed at k , there are $S(n, n - k)$ possible placements on λ .

Proof. Can we convert a set partition into a rook placement and vice versa?

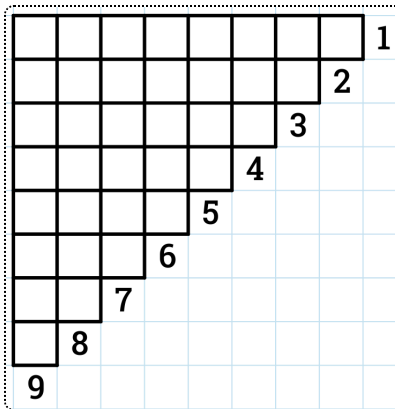
We will demonstrate an example for $n = 9$ and $k = 4$. Let our partition be

$$\pi = (1, 3, 4, 6 \mid 2, 9 \mid 5 \mid 6, 8).$$

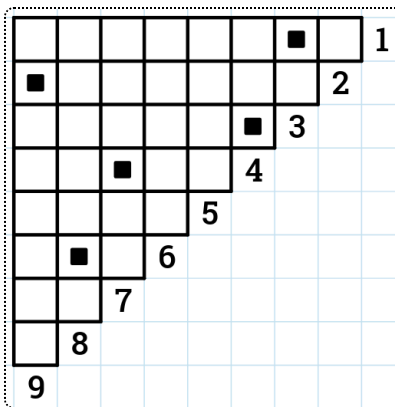
We will represent this permutation in terms of an arc diagram, where we draw a path with arcs adjacent numbers in each block. Note that there is no arc from 5.



Next, we will label our Young diagram with numbers 1 to n . Each label will correspond to one row and column (except 1 has no column and n has no row).



Then, if we have an arc i, j , we place a rook in row i and column j :



Since each number can only be the left endpoint of one arc and the right endpoint of one arc, there is at most one rook in every row and column. Meanwhile, the rooks are always on the board since $i < j$, and there are exactly $n - k$ rooks (since $n - k$ numbers will be the left endpoint of an arc).

The reverse map is straightforward: we can reconstruct the arc diagram and then the underlying set partition. So, there is a bijection and we are done.

Spread-out set partitions

Definition (Spread-out set partitions).

A **spread-out** set partition π of $[n]$ is a set partition such that for all i , i and $i + 1$ are not in the same block of π .

We denote the number of spread-out set partitions of $[n]$ by $\tilde{B}(n)$.

We can once again compute small values:

n	1	2	3	4	5
$B(n)$	1	2	5	15	52
$\tilde{B}(n)$	1	1	2	5	?

It turns out $\tilde{B}(5) = 15$. This suggests our next theorem:

Theorem 4 ($\tilde{B}(n)$).

$$\tilde{B}(n) = B(n - 1) \text{ for } n \geq 1.$$

Proof. Note that under our bijection from the previous section, the restriction of being spread-out is equivalent to saying there is no arc between i and $i + 1$, i.e. there is no rook at $(i, i + 1)$. The positions $(i, i + 1)$ are exactly the ones on the largest antidiagonal of λ , so the number of spread-out set partitions is exactly the number of rook placements on a slightly smaller staircase shape. Using our bijection again, this is $B(n - 1)$, as desired.

Classes of set partitions

For next class, we will define two different classes of set partitions:

- A set partition is called **non-crossing** if its arc diagram has no crossing arcs.
- Similarly, a set partition is **non-nesting** if its arc diagram has no nested arcs.

In our above example, the arcs $(1, 3)$ and $(2, 9)$ cross, while $(6, 8)$ is nested within $(2, 9)$.

Here is another table of small values.

n	1	2	3	4	5
$B(n)$	1	2	5	15	52
# non-crossing	1	2	5	14	?
# non-nesting	1	2	5	14	?

This reminds us of the Catalan numbers, and indeed:

Theorem 5 (Catalan numbers again!).

The numbers of non-crossing and non-nesting set partitions of $[n]$ are both $C(n)$.

We proved something similar in the first lecture, but those were diagrams on $2n$ vertices where each one had degree exactly 1. Try to prove this bijectively as an exercise!

Lecture 13 (March 3)

Ilani is teaching today and Wednesday, and Dora is teaching on Friday.

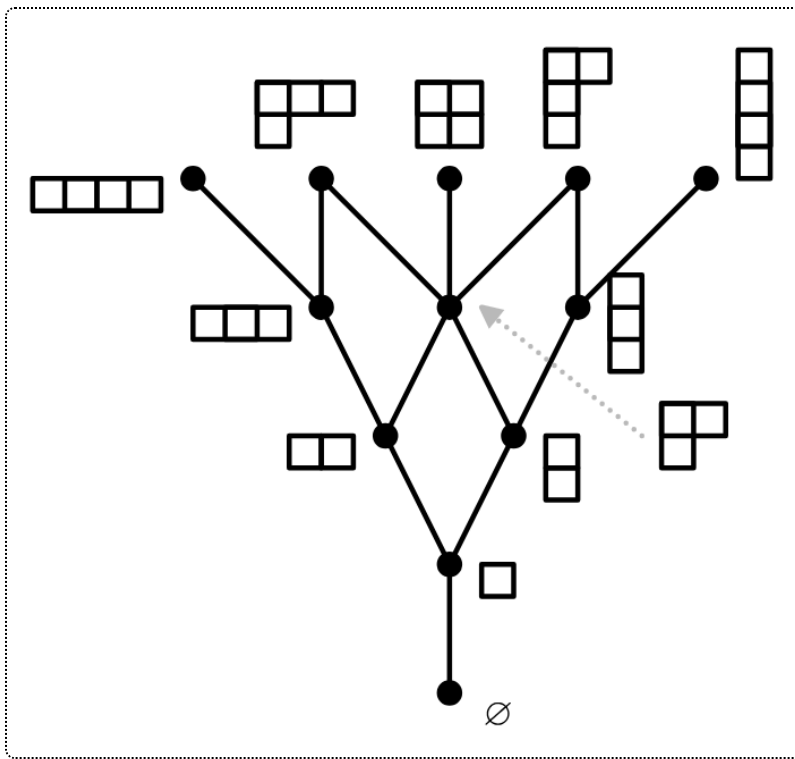
Today we will talk about oscillating tableaux. We will first need some background information:

Linear space on Young's lattice

Definition ($\mathbb{Y}$).

Young's lattice $\mathbb{Y}$ is a poset on Young diagrams where $\lambda \leq \mu$ if we can obtain μ by adding one box to λ .

Here are the first few layers:



Claim 6 (Saturated chains in $\mathbb{Y}$).

Each saturated chain from $\emptyset$ to some λ corresponds to a SYT of λ .

The reason is that we just number the boxes we add in order (similar to the Q of SYT).

Now consider $\mathbb{R}[\mathbb{Y}]$, the linear space (vector space) of formal linear combinations of Young diagrams, so that it has a linear basis given by Young diagrams (we treat each diagram as a linearly independent basis vector). ("Formal" means that we essentially just treat each diagram as a symbol, with no underlying algebra.)

For example, one element of $\mathbb{R}[\mathbb{Y}]$ is the following:

$$5 \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + 22 \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \square & \\ \hline \end{array} - 17 \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} + 6 \emptyset$$

Now, we define two linear operators U and D on $\mathbb{R}[\mathbb{Y}]$ such that:

$$U: \lambda \mapsto \sum_{\mu: \lambda \triangleleft \mu} \mu$$

$$D: \lambda \mapsto \sum_{\mu: \mu \triangleleft \lambda} \mu.$$

(Since they are linear, $U(a\lambda + b\mu) = aU(\lambda) + bU(\mu)$ and the analogous holds for D .)

Essentially, $U(\lambda)$ yields the Young diagrams that are "up" from λ in the poset, and $D(\lambda)$ yields the ones that are "down".

For example:

$$U(\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array}) = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \square & & \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \\ \hline \square & & \\ \hline \end{array} + \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & & & \\ \hline \square & & & \\ \hline \end{array}$$

$$D(\emptyset) = 0 \quad (\text{since nothing is below } \emptyset)$$

$$U(0) = 0 \quad (\text{by linearity})$$

Please note that $\emptyset$ and 0 are different in our vector space! $\emptyset$ is just another basis vector of $\mathbb{R}[\mathbb{Y}]$ that represents an empty diagram, while 0 is the true zero vector (that you get from multiplying the scalar 0 by any vector, or from subtracting any vector from itself.) It makes the algebra more convenient if we separate these two.

(From here I will denote Young diagrams as elements of $\mathbb{R}[\mathbb{Y}]$ by their partition representations colored in green; in lecture they were depicted as actual diagrams, but it's too time-consuming for me to draw those every time. For example, $(1) = \square$. My apologies.)

We can think of this as all possibilities for going "up" or "down" from λ , and we can apply these operators multiple times. For example:

$$\begin{aligned} UU((1)) &= U((1, 1) + (2)) \\ &= ((1, 1, 1) + (2, 1)) + ((2, 1) + (3)) \\ &= (1, 1, 1) + 2(2, 1) + (3) \end{aligned}$$

$$DD((1)) = D(\emptyset) = 0.$$

If we start with λ and apply d down steps and u up steps, we always get 0 if $d - u > |\lambda|$ (since you run out of boxes).

In general, we can see that

$$U^n(\emptyset) = \sum_{\lambda \vdash n} f_\lambda \cdot \lambda,$$

$$D^n(\lambda) = f_\lambda \cdot \emptyset.$$

Recall that f_λ is the number of SYTs of λ , i.e. the number of saturated chains from $\emptyset$ to λ in $\mathbb{Y}$.

$$\text{So, } D^n U^n(\emptyset) = \sum_{\lambda \vdash n} (f_\lambda)^2 \cdot \emptyset.$$

We can think of e.g. $D^3 U^3(\emptyset)$ as counting paths in $\mathbb{Y}$ that start at $\emptyset$, make three "up steps", then three "down steps". By doing casework on the highest tableau λ (with three boxes) that we reach, we see how there are $1 \cdot 1 + 2 \cdot 2 + 1 \cdot 1 = 6$ of these: we have f_λ choices for the upwards path and f_λ choices for the downward path.

Recall from our RSK bijection that $\sum_{\lambda \vdash n} (f_\lambda)^2$ is just $n!$. We can actually see this relation $D^n U^n(\emptyset) = n! \cdot \emptyset$ more directly, using the following crucial lemma:

Lemma 7 (Properties of D and U).

1. $DU - UD = I$ (the identity operator $I(\lambda) = \lambda$ for all λ)
2. $D(\emptyset) = 0$. (We mentioned this earlier.)

For example, if $\lambda = (3, 2)$, $U(\lambda)$ is the sum of the options we get from adding a box, i.e. $(4, 2) + (3, 3) + (3, 2, 1)$. Then $DU(\lambda)$ is

$$\begin{aligned} & (3, 2) + (4, 1) \\ & + (3, 2) \\ & + (2, 2, 1) + (3, 1, 1) + (3, 2). \end{aligned}$$

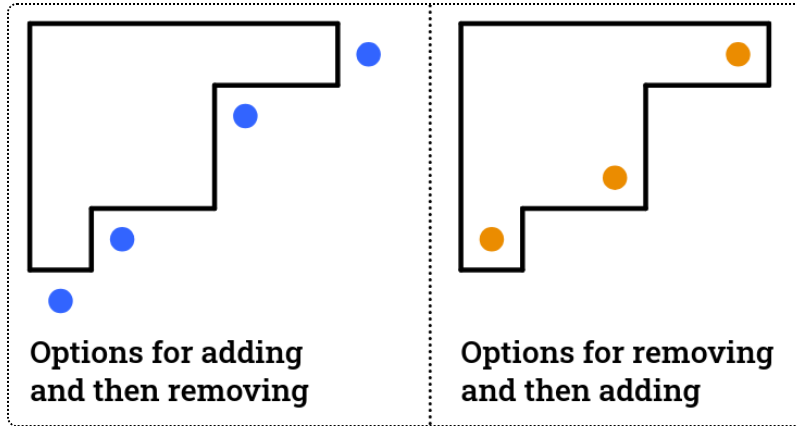
Meanwhile, $D(\lambda)$ is $(2, 2) + (3, 1)$, and

$$\begin{aligned} U(D(\lambda)) &= (3, 2) + (2, 2, 1) \\ &+ (4, 1) + (3, 2) + (3, 1, 1), \end{aligned}$$

which is indeed $DU(\lambda) - I(\lambda)$.

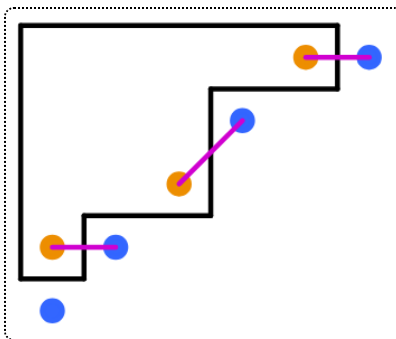
This is not a coincidence: if $\mu \neq \lambda$ and they differ by two boxes (i.e. there's some ν where $\nu \triangleleft \mu$ and $\nu \triangleleft \lambda$), there's exactly one way to remove a box then add a box to go from λ to μ , and the same is true of adding and then removing. (We can think of this operation as "swapping" a corner box for another potential corner.)

However, if we consider the paths that go from λ back to λ , we will need to add and remove the same box in some order. Here are our options:



If we add first, we must use an "outer corner", while if we remove first, we do so at an "inner corner". We can see that there is exactly one more outer corner than inner corner, yielding our identity.

For example, we can pair up all but the bottommost outer corner by matching corners that border the same vertical segment of the right edge, so there must be one extra outer corner.



Now, using this lemma, we can split up $D^n U^n$ into multiple terms:

$$\begin{aligned}
 D^n U^n(\varnothing) &= D^{n-1}(DU)U^{n-1}(\varnothing) \\
 &= D^{n-1}(I + UD)U^{n-1}(\varnothing) \\
 &= D^{n-1}U^{n-1}(\varnothing) + D^{n-1}UDU^{n-1}(\varnothing)
 \end{aligned}$$

If we keep doing this, at the end, we'll get a bunch of terms of the form $I(\emptyset)$ and a bunch of terms of the form (whatever) $D(\emptyset)$. The second type of term vanishes (as $D(\emptyset) = 0$), and there ends up being $n!$ of the first term.

For example:

$$\begin{aligned}
 D^2U^2(\emptyset) &= DU(\emptyset) + DUDU(\emptyset) \\
 &= (I(\emptyset) + UD(\emptyset)) + (DU(\emptyset) + DUUD(\emptyset)) \\
 &= I(\emptyset) + 0 + (I(\emptyset) + UD(\emptyset)) + 0 \\
 &= 2I(\emptyset) + 0 \\
 &= 2\emptyset.
 \end{aligned}$$

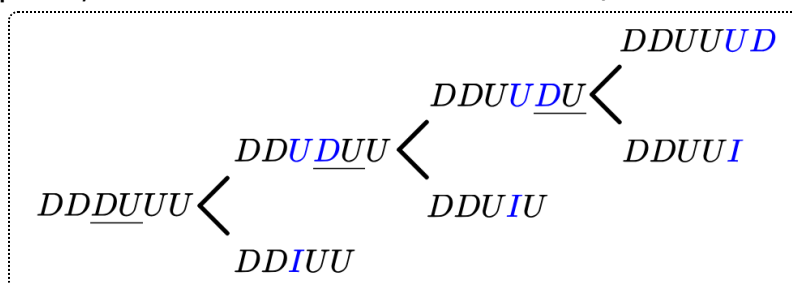
$$\begin{aligned}
 D^3U^3(\emptyset) &= D^2U^2(\emptyset) + D^2UDU^2(\emptyset) \\
 &= 2D^2U^2(\emptyset) + D^2U^2DU(\emptyset) \\
 &= 3D^2U^2(\emptyset) + D^2U^3D(\emptyset) \\
 &= 3D^2U^2(\emptyset) = 6\emptyset.
 \end{aligned}$$

We will elaborate on this next lecture.

Lecture 14 (March 5)

Recall that we are trying to prove algebraically $D^nU^n(\emptyset) = n!\emptyset$. This means that there are $n!$ paths from $\emptyset \in \mathbb{Y}$ that go up n times, go down n times, and return to $\emptyset$. We repeatedly use the fact that $DU = UD + I$.

Explicitly: we create a binary tree starting rooted at e.g. D^nU^n , where each node is the sum of its descendants (obtained by applying $DU = UD + I$). For example, here's what part of a tree could look like: (underlines indicate where the lemma is applied, and blue indicates the results)

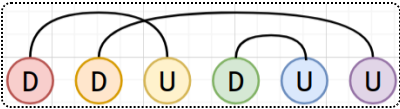


It is possible to continue this until each leaf is either U^iD^i for some $i \geq 1$ or I . Note that $U^iD^i(\emptyset) = 0$.

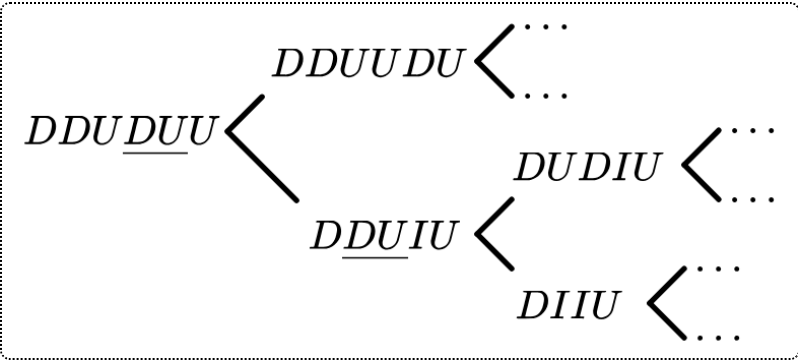
No matter how we specifically expand the tree (we could e.g. apply the lemma to $DDU\underline{DUU}$ instead of $DDU\underline{UDU}$), we will always end up with the same leaves. This is because each path in the tree essentially encodes the information of which pairs of (D, U) , where the D precedes the U , are turned into the identity.

Each instance of I as a leaf corresponds to a perfect matching between D 's and U 's (since none are left over at the end), and there are $n!$ such matchings for D^nU^n , so we get that $D^nU^n(\varnothing) = n!\varnothing$.

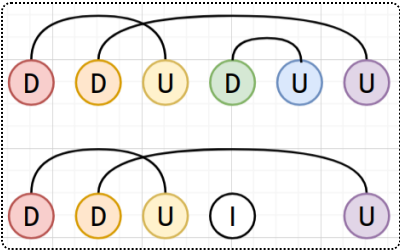
This correspondence between matchings and tree paths is very important. Here's an example: consider the following matching (we color each individual D or U to emphasize that we treat each as an individual object).



Suppose our tree from $DDUDUU$ looks like this.

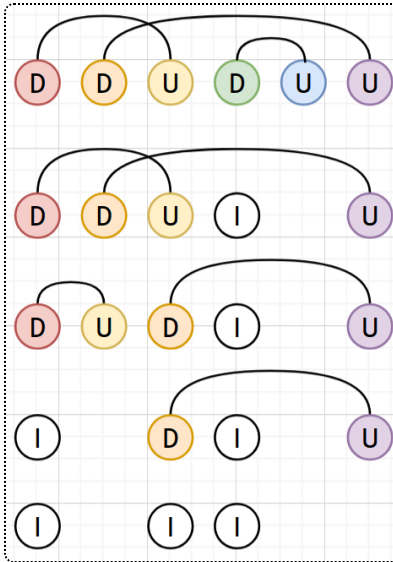


At the first step, we apply the lemma to the green D and blue U . Since they are paired in our matching, we replace with them with I and take the bottom branch.



At the next step, the tree has us apply the lemma to the orange D and the yellow U . Since they are not paired, we take the top branch, where they are

swapped (into UD) rather than replaced with I . Continuing this process may look something like this:



Evidently, no matter what tree we start with, this process always leads us down exactly one path for any given matching.

Theorem 8 (Sum over all arrangements of U 's and D 's).

Let $\mathcal{P}$ denote the set of strings with n U 's and n D 's. Then

$$\sum_{P \in \mathcal{P}} P(\emptyset) = (\# \text{ perfect matchings on } 2n \text{ elements}) \cdot \emptyset$$

(This number equals $(2n - 1)!! = (2n - 1)(2n - 3) \dots (1).$)

Proof. If P is not an (reversed) Dyck path (i.e. some suffix of it has more down steps than up steps), then $P(\emptyset) = 0$ since eventually all of our possibilities disappear.

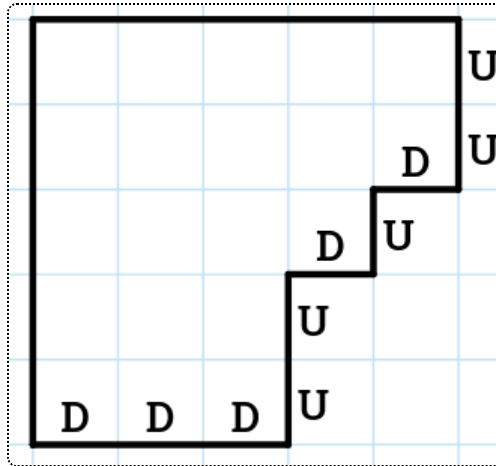
Otherwise, given any string P with n U 's and D 's, each term of $I\emptyset$ in $P(\emptyset)$ corresponds to a matching between D 's and U 's as described above. These matchings are unique: we can recover the string P from any perfect matching since the first vertex in each pair must be D and the second vertex must be U .

Bijection with rook placements

We can also relate these D - U matchings to perfect rook placements. In particular, given a string P of n D s and U s, we can create a unique Young diagram bounded by P : starting from the bottom-left corner of a $n \times n$ grid, we draw the boundary of the diagram by

- moving right for each D , and
- moving up for each U .

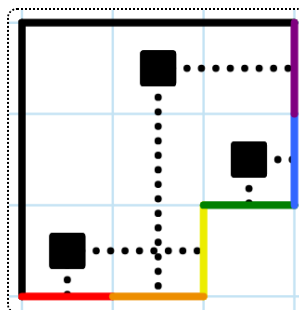
As an example, given $P = DDDUUDUDUU$, we obtain this diagram:



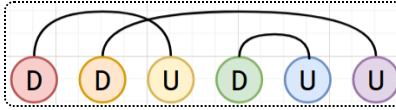
Then, each rook exactly corresponds to one of our D - U pairs from before by taking its row and column, and we can verify that valid placements of n rooks are exactly those that correspond to our desired perfect matchings:

- one rook in each row/column $\iff$ each D/U is paired once
- the rooks are within the boundary $\iff$ each D comes before its paired U .

Our $DDUDUU$ matching from before corresponds to the following (same colors):



You can imagine how if you "unwrap" the boundary and lay it flat, you end up with the same arc diagram as before.



Therefore, we have the following:

Corollary 9 (Rook placements bounded by P).

For a string P , $P(\emptyset) = n\emptyset$, where n the number of rook placements in the Young tableau bounded by P .

Corollary 10 (Perfect matchings).

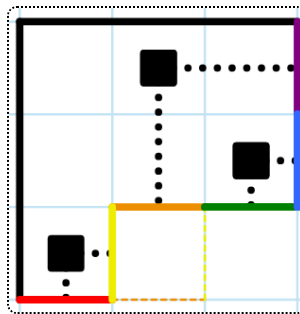
We have a bijection between perfect matchings of $2n$ items and

$\{\text{rook placements bounded by } P \mid P \text{ is a (reversed) Dyck path of length } 2n\}$.

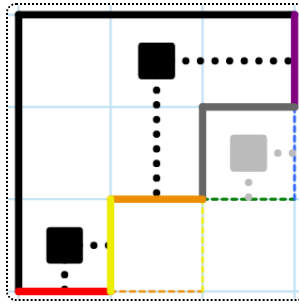
We can actually envision our process of taking paths down the tree as "folding in" corners of this grid. We need to make some slight modifications:

- Replace DU with II instead of I (to signify that it replaced two operators).
- Manually commute I with other operators instead of just skipping over it.

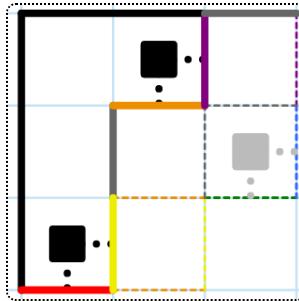
This is what replacing DU with UD looks like (note how yellow and orange swapped positions; $DDUDUU \rightarrow DUDDUU$):



This is what replacing DU with I (really II) looks like (recall that this happens exactly the row/column are matched, i.e. there's a rook):



Now we have $DUDIUI$. If we wanted now to apply the lemma to the orange D and purple U , we would need to first make them adjacent by moving the two I s aside (yielding $DUIDUI$), like so:



We can continue this process until we fold in every square, yielding only a series of I s.

Back to $\mathbb{Y}$

Suppose $P = DDUDDUUU$, as above. Corollary 9 relates $P(\varnothing)$ with rook placements on the tableau bounded by it. But recall that $P(\varnothing)$, by definition, counts paths in $\mathbb{Y}$ that start at $\varnothing$, then go up, up, down, down, up, down, and down (reversed since we apply operators from right to left). Can we associate each individual such path with a rook placement in a natural way?

Indeed, there is, using the same "folding in" process as above!

Theorem 11 (Paths in $\mathbb{Y}$ and rook placements).

Length- $2n$ paths in $\mathbb{Y}$ that starts and end at $\emptyset$ correspond nicely to

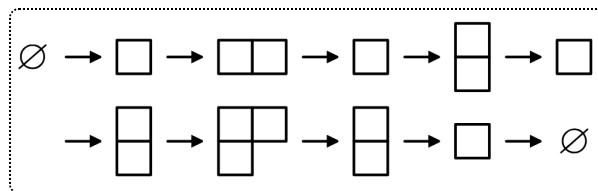
$$\{\text{rook placements bounded by } P \mid P \text{ is a Dyck path of length } 2n\},$$

where P represents the pattern of up and down steps in each path.

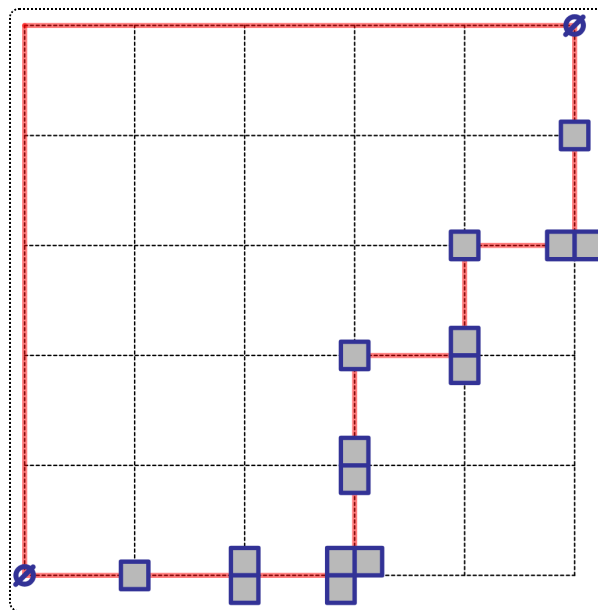
The key idea is that since we've already associated each edge to one of D , U , or I , we might as well make them represent individual covering relations in the poset by labeling each vertex with a Young diagram. Then, we will be able to read off our path by just reading the vertices along the boundary.

Specifically, since we apply the rightmost linear operator first, our path will start from $\emptyset$ in the top-right corner and end at $\emptyset$ in the bottom-left corner. Each time we move down along the boundary, that represents a U , so we add a square to the Young diagram. Every time we move left, that represents a D , so we delete a square. (Note that given a path in $\mathbb{Y}$, this enforces what the boundary P will be, based on the pattern of adding/removing squares).

For example, consider the below path in $\mathbb{Y}$.



We immediately recover the boundary:

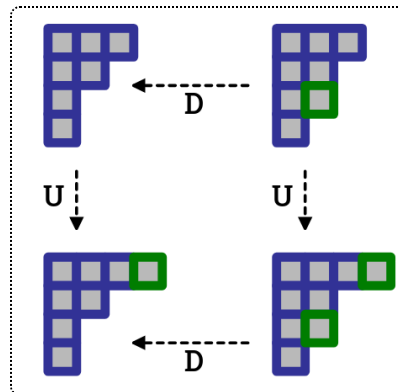


The operations we used to "fold in" the path in the last section were:

- replacing DU with UD ,
- replacing DU with II (corresponding to rooks)
- letting I commute.

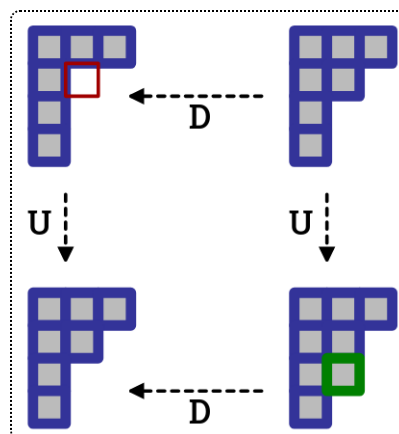
If we want to "fold in" the path here, we need to know what the top-left vertex of a square looks like based on the other three corners. There are only a few possible patterns, based on the current path $\lambda \rightarrow \mu \rightarrow \nu$ (top-right to bottom-right to bottom-left), so we can systematically do this:

(1) If all of λ, μ, ν are distinct, λ and ν differ by exactly two boxes, and as discussed last class, there is only one D - U path and one U - D path between them, so the top-left corner is fixed.



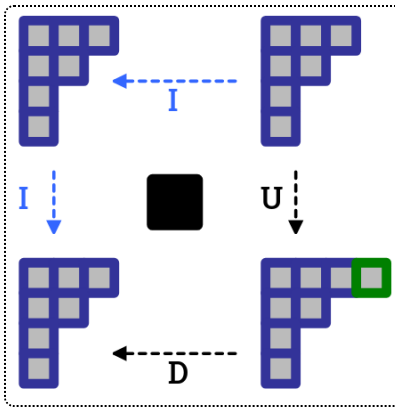
What if $\lambda = \nu$? Recall that $DU = UD + I$, so we need to reserve one case where we place λ into the top-left corner, and then match the remaining options for the top-left with those for the bottom-right (similar to our corner-pairing from last class). One convenient way is to pair the case where we add a box in row $i + 1$ with the one where we remove a box in row i , like this:

(2) Bottom-left and top-right corners are the same; all edges are D or U



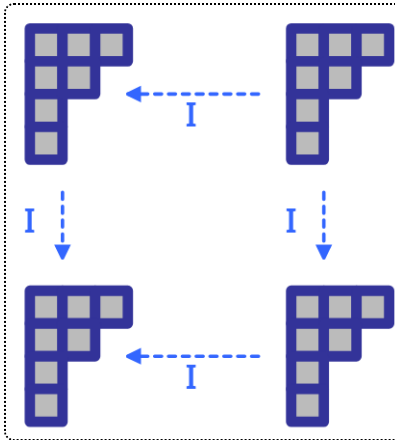
Here, $\lambda \rightarrow \mu$ adds a box in Row 3, so we remove a box in Row 2 for the top-left.

(3) The remaining case is when $\lambda \rightarrow \mu$ adds a box in the first row. In this case, we replace DU with II , so we label the top-left with λ and place a rook in the square (since we're canceling out the U and D , like previously).

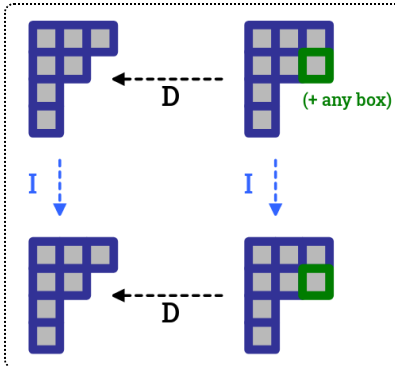


There are two more cases, corresponding to letting I s commute (either with each other or with D/U):

(4) All edges are I :

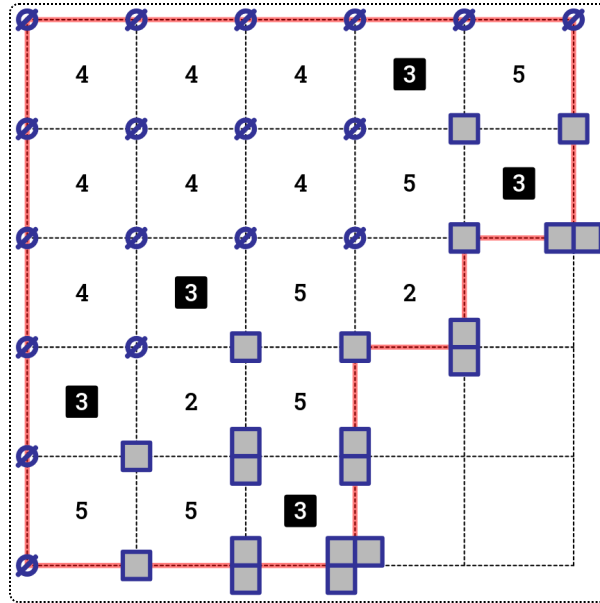


(5) Two opposite edges are I (can be the example shown or its reflection):



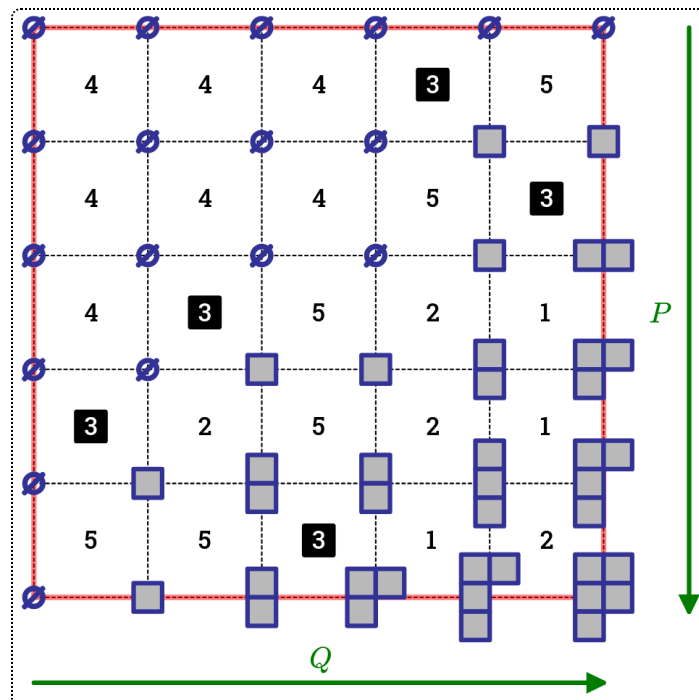
Note that this process of filling out the grid by repeatedly "folding in" the corners of λ_0 is analogous to our process of following a path down the tree from before: when we reach a rook (one of our (D, U) pairs), we replace the U and D with I . and when we don't reach a rook, we simply swap the order from $U \rightarrow D$ to $D \rightarrow U$.

Now, we can fill out all of our vertices in sequence! Each box here is labeled by the pattern it falls under, and our rooks are the squares from pattern 3.

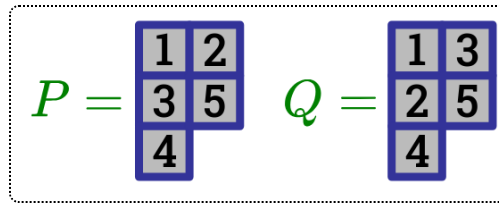


Note that the reverse operation is possible: we know that vertices on the top and left edges must all be $\emptyset$ (as we get rid of all U s and D s), and if we know whether a square has a rook or not, we can determine its bottom-right corner from the other three. So, this is indeed a bijection.

Something interesting happens if we use this to fill in the entire grid and look at the bottom/right edges: we get obtain two different ways to build a certain Young diagram.



We can then make Young tableaux by numbering the squares based on when they were added in each path:



In fact, if we interpret the rook placement as a permutation of columns (in this case, $w = 43512$), this pair (P, Q) is exactly the same as you would get from RSK! This fact also proves the property that w^{-1} maps to (Q, P) in RSK (by reflection across the main diagonal), which we had mentioned earlier but not proved.

Lecture 15 (March 7)

Dora, one of Professor Postnikov's grad students, is teaching about a connection between Young diagrams and topology today. This material will be interesting but not necessarily related to future classes.

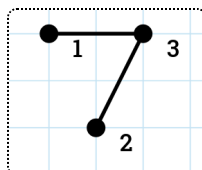
Abstract simplicial complexes

Definition (ASCs).

Let V be a finite set. An **abstract simplicial complex** (ASC) Δ on V is a family of that if subsets of V such that if $\sigma \in \Delta$ and $\tau \subseteq \sigma$, then $\tau \in \Delta$.

For example, if $V = \{1, 2, 3\}$, we could have $\Delta = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 3\}, \{2, 3\}\}$.

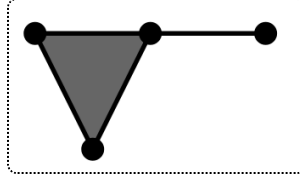
You can geometrically imagine this as a bunch of simplices ("triangles" of various dimensions) glued together, with each element as a vertex, each two-element subset an edge, each three-element subset a triangle, each four-element subset a tetrahedron, etc.



We call each element $\sigma \in \Delta$ a *face*, and the *dimension* of σ is $|\sigma| - 1$.

If $\sigma \in \Delta$ is not contained within some other face of Δ , it is called a *facet*. In our above example, $\{1, 3\}$ and $\{2, 3\}$ are facets.

Δ is called *pure* if all of the facets are of the same dimension. Here is an impure ASC (which has one facet of dimension 2 and one of dimension 1):

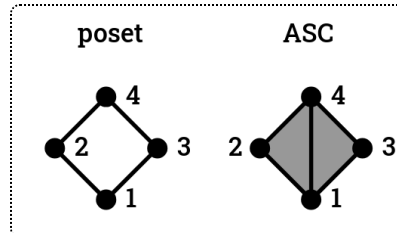


Definition (Complexes from posets).

Given a poset $\mathcal{P}$, we can define the complex $\Delta(\mathcal{P})$ as follows:

- The vertices are elements of $\mathcal{P}$
- Faces are chains in $\mathcal{P}$.

For example, this poset corresponds to an ASC with two dimension-2 facets:

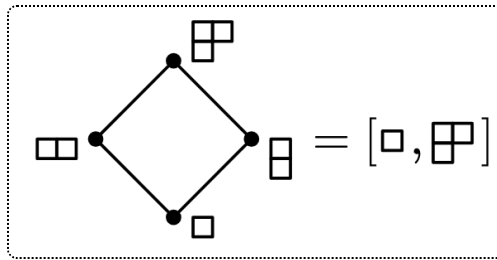


Meanwhile, the poset that comprises one chain of 4 elements corresponds to an ASC that just looks like a tetrahedron (one dimension-3 facet), since any subset will be a chain.

Connection with Young diagrams

The question we want to answer is: what does the ASC corresponding to an interval $[\lambda_1, \lambda_2]$ of $\mathbb{Y}$ look like? (An interval $[u, v]$ of $\mathcal{P}$, where $u \leq v$, comprises all elements between u and v).

For example:



Theorem 12 (Complexes of intervals).

$\Delta([\lambda_1, \lambda_2])$ is homeomorphic to a ball. (e.g. the square-looking ASC above can be deformed into a 2D ball, and a tetrahedron can be deformed into a 3D ball.)

To prove this, we need to consider something called "shellability".

Let $\sigma \in \Delta$. We denote the subcomplex comprising all subsets of σ by $\langle \sigma \rangle$.

Definition (Shelling order).

A **shelling order** of the facets $F_1, F_2, \dots, F_k$ of Δ satisfies that for all j ,

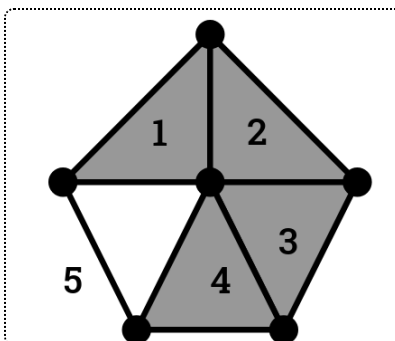
$$\left(\bigcup_{i < j} \langle F_i \rangle \right) \cap \langle F_j \rangle$$

is pure, with all facets being of dimension $|\dim(F_j)| - 1$.

If a complex has a shelling order, it is **shellable**.

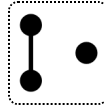
(Useful reference: ["Poset Topology" by Michelle Wachs](#), pp. 41-47)

As an example, here is a shelling order of an ASC:

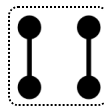


- $\langle F_1 \rangle \cap \langle F_2 \rangle$ is a line (dimension 1 as desired)
- $(\langle F_1 \rangle \cup \langle F_2 \rangle) \cap \langle F_3 \rangle$ is a line (dimension 1 as desired)
- $(\langle F_1 \rangle \cup \langle F_2 \rangle \cup \langle F_3 \rangle) \cap \langle F_4 \rangle$ is a line (dimension 1 as desired)
- $(\langle F_1 \rangle \cup \langle F_2 \rangle \cup \langle F_3 \rangle \cup \langle F_4 \rangle) \cap \langle F_5 \rangle$ is two points (dimension 0 as desired)

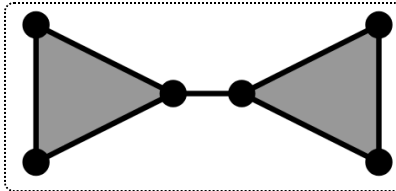
The following complex is shellable, with F_1 on the left and F_2 on the right, since they intersect at $\emptyset$ (dimension -1) as desired.



The following complex is not shellable.



You can also verify that the following complex is not shellable, by looking at all of the possible orderings.



Theorem 13 (Shellable complexes).

A shellable complex is "equivalent" (homotopy type) to a bunch of hollow spheres (of any dimension) glued at a point.

For example, our first example above is equivalent to a circle, since all of the filled triangles can be contracted down (there is just one 2D "hole"). The second example is equivalent to a "1D sphere", the set of points in 1D space that are a certain distance from the origin (i.e. two dots).

The idea involves "homology facets": facets F_j whose entire boundary are contained in the union $\bigcup_{i < j} F_i$. For example, in the above example, the boundary of F_5 is two points which are each in a previous facet. We can rearrange the shelling order so that the homology facets come last. Then, all of the non-homology facets can be contracted down to one point, while the

homology facets become spheres (or circles, or hyperspheres, etc) attached to the point, since the entire boundary maps to that point.

("Boundary" is hard to rigorously define, but is what you intuitively think it means. Imagine a 2D homology facet, i.e. a triangle whose edges all contract down to one point: we can see how the surface closes and becomes a sphere.)

EL-labelings

An "EL-labeling" (edge-lexicographic labeling) of a poset $\mathcal{P}$ is a labeling of its edges such that:

1. In every interval, there is a unique chain where the edge labels are in increasing order. (This is the main condition)
2. The increasing chain is lexicographically minimal.

We have the following nice properties:

Theorem 14 (EL-labeling and shellability).

If $\mathcal{P}$ has an EL-labeling, then the lexicographical ordering of its maximal chains is a shelling order of $\Delta(\mathcal{P})$.

This is (probably) proved in ["Shellable and Cohen-Macaulay Partially Ordered Sets" by Anders Björner](#).

Theorem 15 ($\Delta([\lambda_1, \lambda_2])$ is shellable).

The interval $[\lambda_1, \lambda_2] \in \mathbb{Y}$ has an EL-labeling: each edge represents adding a square to a Young tableau, so label it with the row of the added square. Thus $\Delta([\lambda_1, \lambda_2])$ is shellable.

Here, the unique increasing chain is the one in which squares are added from the top down.

For another example, the Boolean lattice is EL-labelable: label the edge (A_1, A_2) by the element in $A_2 \setminus A_1$. The unique increasing chain is the one where

elements are added in increasing order.

Theorem 16 (Finishing).

If $\Delta(\mathcal{P})$ is shellable, pure, and subthin, then Δ is equivalent to a ball.
(Subthin means that any length-2 interval of $\mathcal{P}$ has at most 4 elements.)

We can see that this applies to any interval $[\lambda_1, \lambda_2]$ of the Young lattice:

- it's pure since all maximal chains are of the same length, and
- it's subthin since an interval of length 2 means you're adding two squares (which there are only two orders for),

so we are done!